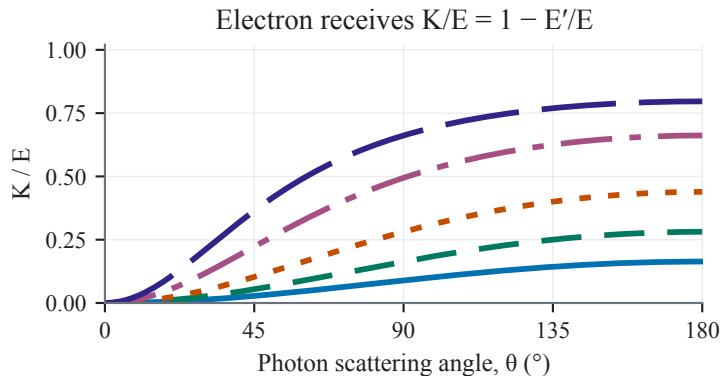
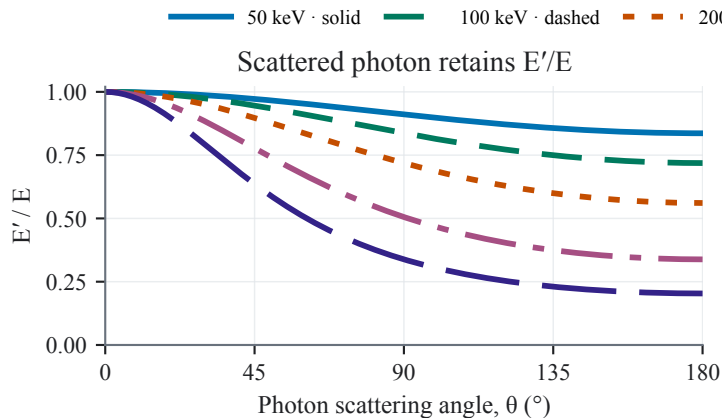


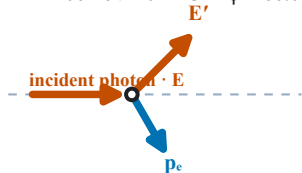
Energy transfer and exact momentum geometry

Exact relativistic free-electron model

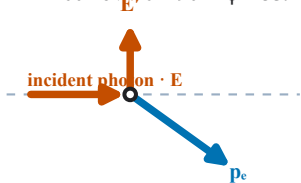
Photon energy is redistributed, never lost: $E = E' + K$ · arrow lengths use one common momentum scale in each collision



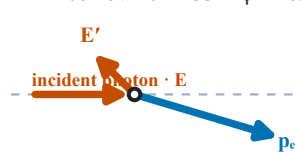
$E = 200 \text{ keV} \cdot \theta = 45^\circ \cdot \phi = 60.0^\circ$



$E = 200 \text{ keV} \cdot \theta = 90^\circ \cdot \phi = 35.7^\circ$



$E = 200 \text{ keV} \cdot \theta = 135^\circ \cdot \phi = 16.6^\circ$



Representative momentum triangles use 200 keV. Electron transverse momentum cancels the scattered photon.